

Reg. No.														
----------	--	--	--	--	--	--	--	--	--	--	--	--	--	--

B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, MAY 2023
First and Second Semester

21CYB101J - CHEMISTRY

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer ALL Questions

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| 1. What is the Geometrical shape of $K_4[Ni(CN)_4]$? | 1 | 1 | 1 | 1 |
| (A) Octahedral | | | | |
| (B) Square planar | | | | |
| (C) Tetrahedral | | | | |
| (D) Trigonal Pyramidal | | | | |
| 2. The crystal field theory considers the metal-ligand bond to be a _____ bond. | 1 | 1 | 1 | 1 |
| (A) Covalent | | | | |
| (B) Ionic | | | | |
| (C) Polar | | | | |
| (D) Hydrogen | | | | |
| 3. The CFSE for a high spin d^4 Octahedral complex is | 1 | 2 | 1 | 2 |
| (A) $-0.6\Delta_{oct}$ | | | | |
| (B) $-1.8\Delta_{oct}$ | | | | |
| (C) $-1.6\Delta_{oct} + P$ | | | | |
| (D) $+1.2\Delta_{oct}$ | | | | |
| 4. In a period with increase in atomic number, the metallic character of an element: | 1 | 2 | 1 | 1 |
| (A) Decreases across period | | | | |
| (B) Increases across period and decrease in group | | | | |
| (C) Increases across period and increase in group | | | | |
| (D) Decreases across period and decrease in group | | | | |
| 5. HASB principle was given by | 1 | 1 | 2 | 1 |
| (A) Lewis | | | | |
| (B) Arrhenius | | | | |
| (C) Bransted | | | | |
| (D) Pearson | | | | |
| 6. Helmholtz free energy A is expressed | 1 | 2 | 2 | 1 |
| (A) $A = U + TS$ | | | | |
| (B) $A = H + TS$ | | | | |
| (C) $A = U - TS$ | | | | |
| (D) $A = H - TS$ | | | | |
| 7. The anode of the galvanic cell has | 1 | 1 | 2 | 1 |
| (A) Positive polarity | | | | |
| (B) Negative polarity | | | | |
| (C) No polarity | | | | |
| (D) Neutral | | | | |

8. In corrosion, as a result of decay, the metals are NOT converted into
 (A) Oxides (B) Hydroxides
 (C) Peroxides (D) Carbonates 1 1 2 1
9. Chiral molecules which are non – super imposable mirror images of each other are called.
 (A) Enantiomers (B) Diastereomers
 (C) Meso compounds (D) Racemic Mixture 1 2 3 1
10. The potential energy of n-butane is minimum for
 (A) Skew conformation (B) Staggered conformation
 (C) Eclipsed conformation (D) Ganche 1 3 3 1
11. Which of the following is an initiator molecule in the free radical polymerisation?
 (A) Benzoyl Peroxide (B) Sulphuric acid
 (C) Potassium permanganate (D) Chromium oxide 1 3 3 1
12. A compound with the same molecular formula exists in two forms one is alcohol and the other is ether, what type of isomerism does it show?
 (A) Metamerism (B) Positional isomerism
 (C) Functional isomerism (D) Chain Isomersim 1 2 3 1
13. The strength of the polymer increases with _____ in molecular weight
 (A) Decreases (B) Increases
 (C) No change (D) Slightly decrease 1 2 4 1
14. Which of the following is NOT a natural polymer?
 (A) Rayon (B) Starch
 (C) Cellulose (D) RNA 1 1 4 1
15. Intermolecular forces of thermoplastic polymers are
 (A) More than elastomers (B) Between elastomers and fibers
 (C) Same as elastomers (D) More than fibers 1 2 4 1
16. Glass transition temperature (T_g) for Nylon – 6:6 is 50°C, which is higher than polyethylene due to
 (A) Vander Waals forces (B) Covalent bonding
 (C) Inter-molecular hydrogen bonding (D) Intra-molecular hydrogen bonding 1 3 4 1
17. Minimum interplanar spacing required for Bragg's diffraction is
 (A) $\frac{\lambda}{4}$ (B) $\frac{\lambda}{2}$
 (C) 4λ (D) 2λ 1 2 5 1
18. The source for XPS is
 (A) Mercury – arc (B) Nernst glower
 (C) Global source (D) Alka 1 2 5 1

- | | |
|---|---|
| 19. What happens in the case when the intermolecular distance increases due to tensile force? | 1 2 5 1 |
| (A) there is no force between the molecules | (B) there seems to be a repulsive force between the molecules |
| (C) there seems to be an attractive force between the molecules | (D) there is zero resultant force between the molecules |
-
- | | |
|--|-------------------|
| 20. Usually stronger constituent of a composite in | 1 1 5 1 |
| (A) Matrix | (B) Reinforcement |
| (C) Both are of equal strength | (D) Can't define |

PART – B (5 × 8 = 40 Marks)

Answer ALL Questions

Marks BL CO PO

- | | |
|--|------------------|
| 21. a.i. Explain briefly about high spin and low spin complexes with examples. | 5 3 1 1 |
| ii. Give the differences between hard and soft acids. | 3 2 1 1 |
| (OR) | |
| b. Write short notes on structural isomerism in coordination compounds. Give examples. | 8 2 1 1 |
-
- | | |
|--|------------------|
| 22. a. With appropriate examples, elucidate how Nernst equation can be applied in a redox reaction and in an acid-base reaction. | 8 3 2 1 |
| (OR) | |
| b. With proper equations compare dry and wet corrosion. | 8 3 2 1 |
-
- | | |
|---|------------------|
| 23. a. Explain Cahn-Ingold prelog priority rules to determine R/S configuration on a chiral center taking an example. | 8 4 3 2 |
| (OR) | |
| b. Sketch the potential energy diagram and explain in detail the conformational analysis of n-butane. | 8 2 3 2 |
-
- | | |
|--|------------------|
| 24. a.i. Give the differences between thermoplastic and thermosets. | 4 1 4 1 |
| ii. How polyurethane is prepared? Give its properties and uses. | 4 1 4 1 |
| (OR) | |
| b. Write a short note on conducting polymer. Explain n and p doping in conducting polymer. | 8 2 4 1 |
-
- | | |
|---|------------------|
| 25. a. Explain Bragg's law with a neat diagram. | 8 2 5 1 |
| (OR) | |
| b.i. Define the terms | 3 1 5 1 |
| 1) Elastic body | |
| 2) Plastic body | |
| 3) Elasticity | |

ii. Write the various engineering applications of composites.

5 1 5 1

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Question

Marks BL CO PO

26. Give a neat sketch of Pourbaix diagram and explain all the significant features.

15 3 2 1

27.i. Explain the stereochemistry of SN1 mechanism.

5 3 3 2

ii. Discuss about the principle and instrumentation of XPS.

10 3 5 1

* * * * *

11. How many optical isomers are possible in a compound with one chiral carbon? (A) 5 (C) 2	(B) 4 (D) 3	1	2	3
12. The potential energy of n-butane is minimum for ----- conformation. (A) Skew (C) Eclipsed	(B) Staggered (D) Gauche	1	2	3
13. In addition polymers, monomers used are ----- . (A) Unsaturated compounds (C) Bifunctional saturated compounds	(B) Saturated compounds (D) Trifunctional saturated compounds	1	1	4
14. The polymer in which the substituents are sterically arranged in a way to give alternate d and l configuration is known as --- . (A) Isotactic polymer (C) Syndiotactic polymer	(B) Atactic polymer (D) Fibres	1	2	4
15. Which of the following is true for the resultant polymer product formed, when molecules of phthalic acid react with molecules of glycerol? (A) Branch polymer (C) Linear polymer	(B) Crosslink polymer (D) Resins	1	2	4
16. Which among following is a naturally occurring polymer? (A) PVC (C) Protein	(B) Acetic acid (D) Polythene	1	1	4
17. Hooke's law essentially defines -----. (A) Elastic limit (C) Strain	(B) Stress (D) Yield point	1	2	5
18. Composites can be classified based on ----- . (A) Matrix type (C) Neither on matrix type nor on reinforcement constituent type	(B) Reinforcement constituent (D) Matrix type and reinforcement constituent	1	1	5
19. ESCA can identify elements in the periodic table above which of the following? (A) Carbon (C) Helium	(B) Boron (D) Potassium	1	2	5
20. With the help of which of the following equations the interplanar distance can be calculated from a known wavelength of the source and measured angle? (A) Coolidge equation (C) Debye equation	(B) Scherrer equation (D) Bragg's equation	1	2	5

PART - B (5 × 8 = 40 Marks)

Marks BL CO

Answer all Questions

21. (a) i. With suitable diagrams, discuss the crystal field splitting in the octahedral complex. (6 Marks) ii. How are the crystal field splitting energies for octahedral (Δ_0) and tetrahedral (Δ_t) complexes related? (2 Marks) (OR) (b) i. Explain briefly the high spin and low spin complexes with examples. (4 Marks) ii. What is the screening effect? Calculate shielding constant and effective nuclear charge for 4s electron in Mn. (4 Marks)	8	2	1
22. (a) Derive Gibb's-Helmoltz equation and explain its applications. (OR) (b) i. Corrosion in an electrochemical phenomenon; explain. (6 Marks) ii. What is the purpose of using salt bridge in Galvanic cell. (2 Marks)	8	2	2

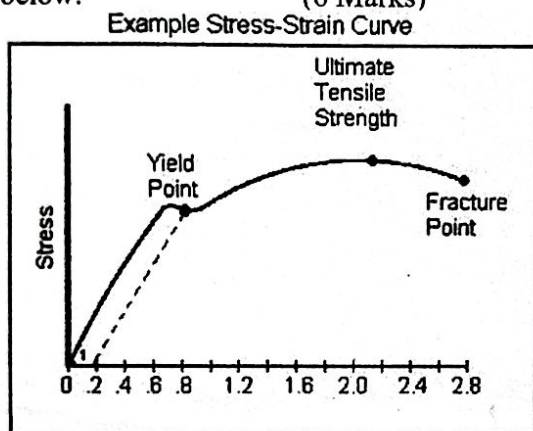
23. (a) i. Explain Cahn-Ingold Prelog rules to determine R/S configuration on a chiral center, with an example (6 Marks) 8 3 3
 ii. Write a note on position isomerism in organic compounds. (2 Marks)

(OR)

- (b) Explain in detail the conformational analysis of n-butane with a potential energy diagram.
24. (a) i. Explain n-doping and p-doping in conducting polymers. (3 Marks) 8 2 4
 ii. What are the differences between Thermoplastic and Thermosets? Give examples. (5 Marks)

(OR)

- (b) i. How are the following plastics synthesized? Give their applications (4 Marks)
 (1) Nylon 6:6 (2) PTFE
 ii. Define the degree of polymerization and functionality of monomer.
25. (a) i. Explain the points given in the stress-strain curve (6 Marks) 8 3 5
 below:



- ii. Compute the Miller Indices for a plane intersecting at $x = \frac{1}{4}$, $y = 1$ and $z = \frac{1}{2}$. (2 Marks)

(OR)

- (b) i. Explain Bragg's law with a neat sketch. How it is applied for studying the diffraction of X-rays by atoms in a crystalline structure? (6 Marks)
 ii. Mention important characteristics of composite material. (2 Marks)

PART - C (1 × 15 = 15 Marks)

Answer any 1 Questions

Marks BL CO

26. i. With a neat sketch, explain the Pourbaix diagram for Iron. (10 Marks) 15 3 3
 ii. Explain the free radical mechanism for addition reaction, with an example (5 Marks)
27. i. Discuss the principle, instrumentation and applications of XPS (10 Marks) 15 3 5
 ii. Calculate the spin-only magnetic moment for following: (5 Marks)
 Fe^{+2} . (atomic number of Fe is 26)
 Cr^{3+} (atomic number of Cr is 24)

* * * * *